

Troubleshooting Guide

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Troubleshooting Guide

Overview

This guide provides solutions for common issues encountered during ClientIQ deployment and operation. Issues are organized by component.

Quick Diagnostics

Health Check Commands

```
# Check all services
Get-Service -Name "ClientIQ", "Nginx" | Format-Table Name, Status, StartType

# Test application health
Invoke-RestMethod -Uri "http://localhost:5000/api/health" -Method GET

# Test via Nginx
Invoke-RestMethod -Uri "https://clientiq.yourbank.com/api/health" -Method GET -SkipCertificateCheck

# Check database connectivity
Invoke-Sqlcmd -ServerInstance "sql-server" -Database "ClientIQ" -Query "SELECT 1"
```

Log Locations

Component	Log Path
Application	C:\ClientIQ\logs\stdout.log
App Errors	C:\ClientIQ\logs\stderr.log
Nginx Access	C:\nginx\logs\clientiq_access.log
Nginx Error	C:\nginx\logs\clientiq_error.log
Windows Events	Event Viewer > Applications

Application Issues

NODE_ENV Not Recognized (Windows)

Symptoms:

'NODE_ENV' is not recognized as an internal or external command,
operable program or batch file.

Cause: Unix-style inline environment variable syntax (`NODE_ENV=development npm run dev`) does not work in Windows Command Prompt or PowerShell.

Solutions:

1. Use PowerShell environment variables:

```
$env:NODE_ENV = "development"  
npm run dev
```

2. Use Command Prompt with set:

```
set NODE_ENV=development && npm run dev
```

3. Create a Windows batch file (dev.bat):

```
@echo off  
set NODE_ENV=development  
set PORT=5000  
set DATABASE_DIALECT=sqlserver  
REM Add other environment variables as needed  
npm run dev
```

See Also: [Dev/Test Server Setup Guide](#) for complete Windows development environment configuration including ready-to-use batch files and PowerShell scripts.

ENOTSUP Error: Operation Not Supported on Socket (Windows)

Symptoms:

```
Error: listen ENOTSUP: operation not supported on socket 0.0.0.0:5000  
code: 'ENOTSUP',  
errno: -4049,  
syscall: 'listen',  
address: '0.0.0.0',  
port: 5000
```

Cause: Windows Server may not support binding to `0.0.0.0` on certain network configurations.

Solution: Set `HOST=127.0.0.1` instead of `0.0.0.0`:

```
# PowerShell  
$env:HOST = "127.0.0.1"
```

```
# Command Prompt
set HOST=127.0.0.1
```

```
# .env file
HOST=127.0.0.1
```

Note: This works because Nginx handles external traffic and proxies to `http://127.0.0.1:5000` on the same server.

Application Won't Start

Symptoms:

- Service fails to start
- Process exits immediately

Diagnostic Steps:

1. Check service status:

```
Get-Service -Name "ClientIQ" | Format-List *
```

2. Review error logs:

```
Get-Content "C:\ClientIQ\logs\stderr.log" -Tail 50
```

3. Test manually:

```
cd C:\ClientIQ\app
$env:DOTENV_CONFIG_PATH = "C:\ClientIQ\config\.env"
node dist/server/index.js
```

Common Causes:

Error	Solution
Cannot find module	Run <code>npm ci --production</code>
EADDRINUSE	Port 5000 in use, stop other service
Missing environment variable	Check <code>.env</code> file exists and readable
Database connection failed	Verify SQL Server connectivity
NODE_ENV not recognized	See "NODE_ENV Not Recognized" section above

High Memory Usage

Symptoms:

- Memory usage over 2GB
- Application becomes unresponsive

Solutions:

1. Set Node.js memory limit:

```
nssm set ClientIQ AppParameters "--max-old-space-size=2048 dist/server/index.js"
nssm restart ClientIQ
```

2. Check for memory leaks:

```
# Monitor memory over time
while ($true) {
    Get-Process -Name "node" | Select-Object WorkingSet64, CPU
    Start-Sleep -Seconds 60
}
```

3. Enable garbage collection logging:

```
# Add to environment
NODE_OPTIONS="--expose-gc --trace-gc"
```

Slow API Responses

Symptoms:

- API calls take > 5 seconds
- Timeout errors

Diagnostic Steps:

1. Check application logs for slow queries:

```
Select-String -Path "C:\ClientIQ\logs\stdout.log" -Pattern "slow|timeout|error" -Context 2,5
```

2. Review Nginx upstream timing:

```
Select-String -Path "C:\nginx\logs\clientiq_access.log" -Pattern "urt=\.([0-9]+\.[0-9]+)" |  
    ForEach-Object { $_.Matches.Groups[1].Value } |  
    Measure-Object -Average -Maximum
```

3. Check database performance:

```
-- Top 10 slowest queries  
SELECT TOP 10  
    qs.total_elapsed_time / qs.execution_count AS avg_ms,  
    SUBSTRING(st.text, 1, 200) AS query_text  
FROM sys.dm_exec_query_stats qs  
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) st  
ORDER BY avg_ms DESC;
```

Nginx Issues

502 Bad Gateway

Cause: Nginx cannot reach the Node.js application.

Solutions:

1. Verify application is running:

```
Test-NetConnection -ComputerName localhost -Port 5000
```

2. Check Nginx upstream configuration:

```
# Verify in nginx.conf  
upstream clientiq_backend {  
    server 127.0.0.1:5000;  
}
```

3. Review Nginx error log:

```
Get-Content "C:\nginx\logs\error.log" -Tail 20
```

504 Gateway Timeout

Cause: Application response takes too long.

Solutions:

- 1. Increase Nginx timeout:

```
location /api/ {
    proxy_read_timeout 120s;
    proxy_connect_timeout 60s;
}
```

- 2. Optimize slow database queries
- 3. Check for blocking operations in code

SSL Certificate Errors

Symptoms:

- Browser shows certificate warning
- ERR_CERT_* errors

Diagnostic:

```
# Check certificate validity
openssl s_client -connect clientiq.yourbank.com:443 -servername clientiq.yourbank.com

# Verify certificate dates
openssl x509 -in "C:\nginx\conf\ssl\clientiq.crt" -noout -dates
```

Solutions:

Error	Solution
Certificate expired	Renew and replace certificate
Name mismatch	Verify CN/SAN matches hostname
Untrusted CA	Install intermediate certificates
Key mismatch	Regenerate CSR with correct key

Database Issues

Connection Refused

Symptoms:

- ECONNREFUSED errors
- "Cannot connect to SQL Server"

Diagnostic:

```
# Test network connectivity
Test-NetConnection -ComputerName sql-server.yourbank.local -Port 1433

# Check SQL Server Browser
Get-Service -ComputerName sql-server.yourbank.local -Name "SQLBrowser"
```

Solutions:

1. Verify SQL Server is running on target host
2. Check firewall allows port 1433
3. Ensure SQL Server is configured for TCP/IP
4. Verify correct server name in connection string

Authentication Failed

Symptoms:

- "Login failed for user"
- Error 18456

Solutions:

1. Verify credentials:

```
-- On SQL Server
SELECT name, is_disabled
FROM sys.sql_logins
WHERE name = 'svc_clientiq';
```

2. Check login permissions:

```
USE ClientIQ;
SELECT dp.name, dp.type_desc, p.permission_name
FROM sys.database_principals dp
JOIN sys.database_permissions p ON dp.principal_id = p.grantee_principal_id
WHERE dp.name = 'svc_clientiq';
```

3. Reset password if needed:

```
ALTER LOGIN svc_clientiq WITH PASSWORD = 'NewPassword123!';
```

Query Timeout

Symptoms:

- Queries exceed 30 second timeout
- "Execution Timeout Expired"

Solutions:

1. Identify slow queries:

```
SELECT TOP 10
    total_elapsed_time / execution_count AS avg_time,
    execution_count,
    SUBSTRING(st.text, 1, 200) AS query
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) st
ORDER BY avg_time DESC;
```

2. Check for missing indexes:

```
SELECT
    'CREATE INDEX [IX_' + OBJECT_NAME(mid.object_id) + '] ON ' +
    mid.statement + ' (' + mid.equality_columns + ')' AS CreateStatement,
    migs.avg_user_impact
FROM sys.dm_db_missing_index_groups mig
JOIN sys.dm_db_missing_index_group_stats migs ON mig.index_group_handle = migs.group_handle
JOIN sys.dm_db_missing_index_details mid ON mig.index_handle = mid.index_handle
WHERE mid.database_id = DB_ID('ClientIQ')
ORDER BY migs.avg_user_impact DESC;
```

3. Check for blocking:

```
SELECT
    blocking_session_id,
    session_id,
    wait_type,
    wait_time / 1000 AS wait_seconds
FROM sys.dm_exec_requests
WHERE blocking_session_id > 0;
```

SAML Authentication Issues

Redirect Loop

Symptoms:

- Browser continuously redirects between app and IdP
- "Too many redirects" error

Solutions:

1. Verify ACS URL matches IdP configuration exactly
2. Check session cookies are being set:

```
// Browser DevTools > Application > Cookies
// Look for clientiq.sid cookie
```


3. Ensure Nginx passes required headers:

```
proxy_set_header X-Forwarded-Proto $scheme;  
proxy_set_header X-Forwarded-Host $host;
```

Invalid Signature

Symptoms:

- "Signature validation failed"
- User cannot complete login

Solutions:

1. Verify IdP certificate is current:

```
# Check certificate not expired  
$cert = New-Object System.Security.Cryptography.X509Certificates.X509Certificate2  
$cert.Import([Convert]::FromBase64String($env:SAML_CERT))  
$cert.NotAfter
```

2. Ensure certificate is properly formatted (no line breaks in env var)

3. Verify signature algorithm matches (SHA-256 required)

Missing Attributes

Symptoms:

- "Missing required SAML attributes" error
- User created without expected data

Solutions:

1. Enable debug logging:

```
DEBUG=passport:*,saml:*
```

2. Review SAML assertion in IdP logs

3. Verify attribute mapping:

```
// Check these URIs match your IdP  
const ATTRIBUTE_MAP = {  
  employeeId: 'http://schemas.xmlsoap.org/ws/2005/05/identity/claims/employeeid',  
  // ... verify each mapping  
};
```

Performance Issues

Slow Page Load

Diagnostic Checklist:

1. Network latency:

```
# From client to server
Test-NetConnection -ComputerName clientiq.yourbank.com -Port 443
```

2. Database response time:

```
SET STATISTICS TIME ON;
SELECT * FROM customer WHERE customer_id = 1;
SET STATISTICS TIME OFF;
```

3. Application metrics:

```
# Check request timing in Nginx logs
Select-String -Path "C:\nginx\logs\clientiq_access.log" -Pattern "rt=([0-9.]+)" -AllMatches |
  ForEach-Object { [double]$_Matches.Groups[1].Value } |
  Measure-Object -Average -Maximum -Minimum
```

Memory Exhaustion

Symptoms:

- "JavaScript heap out of memory"
- Application crashes under load

Solutions:

1. Increase Node.js heap size:

```
nssm set ClientIQ AppParameters "--max-old-space-size=4096 dist/server/index.js"
```

2. Enable memory profiling:

```
NODE_OPTIONS="--inspect --expose-gc"
```

3. Use Chrome DevTools for memory analysis:

- Navigate to `chrome://inspect`
 - Connect to Node.js process
 - Take heap snapshots
-

Log Analysis

Common Log Patterns

Successful authentication:

```
[SAML] Processing authentication for profile: { nameID: "user@bank.com" }  
[SAML ACS] Session created: { employeeId: 123, roles: ["Teller"] }
```

Database connection:

```
[Database] Detected dialect: mssql  
[Database] Connection pool initialized
```

Error patterns to watch:

```
[ERROR] Database connection failed: ECONNREFUSED  
[SAML] Signature validation failed  
[API] Request timeout after 30000ms
```

Log Aggregation

```
# Aggregate errors by type  
Get-Content "C:\ClientIQ\logs\stderr.log" |  
  Select-String -Pattern "\[(ERROR|WARN)\]" |  
  Group-Object { $_.Matches.Groups[1].Value } |  
  Sort-Object Count -Descending
```

Emergency Procedures

Application Unresponsive

```
# 1. Restart application  
Restart-Service ClientIQ -Force  
  
# 2. If still unresponsive, kill process  
Get-Process -Name "node" | Stop-Process -Force  
Start-Service ClientIQ  
  
# 3. Check for resource exhaustion  
Get-Counter '\Memory\Available MBytes', '\Processor(_Total)\% Processor Time'
```

Database Emergency

```
-- Kill blocking sessions
DECLARE @kill varchar(8000) = '';
SELECT @kill = @kill + 'KILL ' + CAST(session_id AS varchar) + '; '
FROM sys.dm_exec_requests
WHERE blocking_session_id = 0
AND session_id IN (SELECT blocking_session_id FROM sys.dm_exec_requests);
EXEC(@kill);
```

Rollback Deployment

```
# 1. Stop services
Stop-Service ClientIQ, Nginx

# 2. Restore previous version
Copy-Item "C:\ClientIQ\backups\app-previous\*" "C:\ClientIQ\app" -Recurse -Force

# 3. Restore configuration
Copy-Item "C:\ClientIQ\backups\config-previous\.env" "C:\ClientIQ\config\.env" -Force

# 4. Start services
Start-Service ClientIQ, Nginx
```

Support Escalation

When to Escalate

- Application down > 15 minutes
- Data corruption suspected
- Security incident
- Performance degradation affecting all users

Information to Gather

1. Error messages (exact text)
2. Timestamps of issue
3. Affected users/features
4. Recent changes
5. Log excerpts

Contact Information

Level	Team	Contact	Response Time
L1	Help Desk	helpdesk@yourbank.com	15 min
L2	App Support	appsupport@yourbank.com	1 hour
L2	DBA Team	dba@yourbank.com	1 hour
L3	Development	dev-oncall@yourbank.com	4 hours

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